

Riemann Roch Generalized

Consider the Gauss-Bonnet theorem ([this](#)):

$$\int_M K dA = \int_{\partial M} k_g ds = 2\pi\chi(M)$$

where M is a 2-dimensional compact Riemann manifold, K the Gaussian curvature, k_g the geodesic curvature, and $\chi(M)$ the Euler characteristic. This can be thought of as a special case of The Riemann-Roch Theorem!

There is also a more direct topological generalization of the Gauss-Bonnet theorem known as the Chern theorem, or Chern–Gauss–Bonnet theorem, to $2n$ -dimensional Riemann manifolds without boundary:

$$\chi(M) = \int_M e(\Omega)$$

where $\chi(M)$ is the Euler characteristic, Ω is the curvature form (or Levi-Civita connection), and $e(\Omega)$ is the Euler class:

$$e(\Omega) = \frac{1}{(2\pi)^n} \text{Pf}(\Omega)$$

where Pf is the *Pfaffian*; namely as Ω can always be represented as a skew-symmetric matrix (when the dimension of M is even, hence the $2n$) it can always be written as the square of a polynomial in matrix entries, namely if A is the skew-symmetric matrix:

$$\text{Pf}(A)^2 = \det(A)$$

for example:

$$A = \begin{pmatrix} 0 & a \\ -a & b \end{pmatrix} \quad \text{Pf}(A) = a$$

$$B = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix} \quad \text{Pf}(B) = af - be + dc$$

Atiyah–Singer index theorem seems to generalize what I’m working on further.

Rapid fire concepts we know:

- We shall take varieties and schemes for granted
- We shall be working with algebraic curves, usually algebraic plane curves; hence curves given by $f(x, y)$ for coordinate functions x, y .
- Such curves are projective ([1]). If they are smooth, we naturally ask about their relation to one-dimension complex manifolds; these are called Riemann surfaces (as they are 2 real dimensional)

- Compact Riemann surfaces are classified up to homeomorphism by their genus, and $\chi(X) = 2 - 2g$ where χ is the Euler Characteristic.
- Via normalization, we have the following (This is in fact proven in [2])

Theorem 0.1: Characterizing Riemann Surfaces Using Normalization

Any compact Riemann surface $\tilde{C}$ can be obtained through the normalization of a certain plane algebraic curve C with at most ordinary double points, i.e. There exists a holomorphic mapping $\sigma : \tilde{C} \rightarrow \mathbb{P}^2(\mathbb{C})$ such that $\sigma(\tilde{C})$ is an algebraic curve possessing at most ordinary double points

- The geometry polynomial function (by FTC) makes them surjective and n -to-1 except at the ramification points.
- We shall take holomorphic and meromorphic differential forms for granted
- Key result about holomorphic differential forms:

Theorem 0.2: Stokes Theorem For Holomorphic Differentials

Let C be a Riemann surface, $\Omega \subseteq C$ an open set, $\bar{\Omega}$ compact and $\partial\Omega = \gamma$ a piecewise smooth curve, and ω a holomorphic differential defined on an open set containing $\bar{\Omega}$. Then:

$$\int_{\partial\Omega} \omega = 0$$

Proof :

use local coordinates and apply Cauchy's theorem to get

$$\int_{\partial\Omega_i} \omega = 0$$

on the components, which then gluing gives the desired result

- Residue is winding number

1 Poincaré-Hopf Formula

Let C be a compact Riemann surface. Then we know that for any nonconstant $f \in K(C)$ (the set of meromorphic functions on C), we get:

$$\sum_{p \in C} \nu_p(f) = 0 \tag{1}$$

that is, the number of zeros plus the number of poles (counting multiplicity) is zero. Phillip actually has this really cool generalization of the proof that I got in [3], he prove the following:

Theorem 1.1: Residue Theorem For Forms

Let C be a compact Riemann surface and $\omega \in K^1(C)$ a differential form (the K^1 is his notation for diff. form). Then:

$$\sum_{p \in C} \text{Res}_p(\omega) = 0$$

Proof :
exercise

Then you can deduce the usual equation (1) by setting $\omega = \frac{df}{f}$. Now, we can ask the follow: for a trivial $\omega \in K^1(C)$, what is:

$$\sum_{p \in C} \nu_p(\omega)$$

A general differential form on a compact Riemann surface has more flexibility than functions, and the order at p is certainly well-defined the same way the residue of a differential form is well defined, so this is a meaningful question. As one-dimensional differential forms are co-vector fields, this is equivalent to a problem on vector-fields, and we know that one interpretation of the Euler-characteristic on surfaces is to count the zero-points (which are sink and source points) and the saddle points (notice how closely this naturally relates to curvature; this is why it comes up naturally in the Gauss-Bonnet Theorem) and so we get:

$$\sum_{p \in C} \nu_p(\omega) = -\chi(C)$$

- There is an interesting word on the schematic representation of curves, [2, p.28]
- Note that meromorphic functions on complex manifold need not be able to map to $\mathbb{CP}^1$, consider z/w from $\mathbb{C}^2$.
- The end of chapter I, section 10, has a very very simple proof on when one can embedd a Riemann surface into $\mathbb{CP}^n$!

2 About Normalization

- He starts with a general plane curve, defines singular points to be when the three partial are zero (hence no linear term when translated to $[0 : 0 : 1]$, or in his notation $[1 : 0 : 0]$). We can then consider any curve $f(x, y) = F(1, x, y)$ when considering the canonical embedding $(x, y) \mapsto [1 : x : y]$. Then it must be double/triple/tuple if the lowest degree is $\geq 2, \geq 3, \geq k$. It is or ordinary if all the tangent line are distinct.
- We can now consider irreducible components. If it is over a UFD, then the GCD is well-defined and so we get that f, g have no common factors if $(f) + (g) = (1)$ so that if $(f) + (g) = (h)$ then $f = Fh$ and $g = Gh$ so that $fG = gF$. We can detect this using the resultant, and hence we can detect singular points over non-characteristic 0 fields using the discriminant. We can then proceed to show that if C is a curve, S the singular points, and $C^* = C \setminus S$ the compliment of the singular points, then C^* is connected, not necessarily compact, S is finite.

- Next comes normalization which he defines as follows:

Definition 2.1: Normalization

Suppose C is a irreducible plane algebraic curve, S the set of its singular points. Then there exists a compact Riemann surface $\tilde{C}$ and a holomorphic mapping

$$\sigma : \tilde{C} \rightarrow \mathbb{CP}^2$$

such that

1. $\sigma(\tilde{C}) = C$
2. $\sigma^{-1}(S)$ is a finite set
3. $\sigma : \tilde{C} \setminus \sigma^{-1}(S) \rightarrow C \setminus S$ is injective

Then $(\tilde{C}, \sigma)$ is the normalization of C , and we write just $\tilde{C}$ if it is unambiguous in context.

- We get the following result equivalent to birational maps:

Lemma 2.2: Dense Injective, then Biholomorphic

Suppose $\tilde{C}, \tilde{C}'$ are Riemann surfaces and

$$h : \tilde{C} \rightarrow \tilde{C}'$$

is a surjective holomorphic mapping which is injective on a dense open subset of $\tilde{C}$. Then h is a biholomorphic mapping.

Proof :

Prove it is biholomorphic in every neighbourhood.

- Then

Theorem 2.3: Uniqueness of Normalization

The normalization of an algebraic curve C is unique up to isomorphism (it commutes in the way you expect)

Proof :

use above lemma.

The goal now is to prove a normalization always exists.

- Next Griffiths defines $\mathbb{C}\{x\}$ which is not quite $\mathbb{C}[[x]]$ because it is the set of all *convergent* power series of the form $\sum_k a_k x^k$. Similarly for $\mathbb{C}\{x, y\}$ which is convergent $\sum_{i,j} a_{ij} x^i y^j$.

Then a Weierstrass polynomial is a polynomial of the form. He denotes the later by $\mathcal{O}$.

$$w = y^d + a_1(x)y^{d-1} + \cdots + a_d(x)$$

where

$$a_j(x) \in \mathbb{C}\{x\} \quad a_j(0) = 0 \quad j = 1, \dots, d$$

Then he goes on to prove the *Weierstrass Propagation theorem* which say that under the appropriate generic conditions, any $f \in \mathcal{O}$ can be written as $f = u \cdot w$ where u is a unit (so $1/u \in \mathcal{O}$ and w is a Weierstrass polynomial.

- Proving this, we get the following great corollary:

Corollary 2.4: Ring Structure of Analytic Func

$\mathcal{O}$ is a UFD, hence any $f \in \mathcal{O}$ can be expressed up to unit u as:

$$f = uf_1f_2 \cdots f_n$$

where each f_i is irreducible in $\mathcal{O}$.

Proof :

Note that proving $\mathbb{C}\{x\}$ is a UFD is easy because any $f(x) = x^m h(x)$ with $h(x)$ a unit with nonzero constant. For the rest checkout [2].

- An important realization is that being irreducible in $\mathbb{C}\{x\}[y]$ and $\mathbb{C}\{x, y\}$ is the same thing.
- Next, let's take an irreducible curve, center the point of interest at $(0, 0)$, represent it as

$$y^n + a_1(x)y^{n-1} + \cdots + a_d(x)$$

Then locally at 0, we may have multiple local irreducible components. Let's take the curve over $\mathbb{C}^2$. Then around 0, if there are multiple components, it will look like multiple disks that are glued together at a point.

- Using this, we can locally normalize, and then use that to globally normalize
- Next Griffiths looks at intersection numbers and Bézout Theorem.

Ramification Divisors and the Riemann-Hurwitz Formula

Let $\nu_f(p)$ be the order of a point of a holomorphic mapping $f : C \rightarrow C'$ between Riemann surfaces. As such a map is always surjective if it is non-constant, we have $\nu_f(p) \geq 1$. It ramifies if $\nu_f(p) > 1$, and as ramified points are isolated we know that $\{p \in C \mid \nu_f(p) > 1\}$ is finite. Thus we have the well-defined:

Definition 2.5: Ramification Divisor

The *ramification divisor* of f is :

$$R = \sum_{p \in C} (\nu_f(p) - 1)p \in \text{Div}(C)$$

Recall that if $D = \sum_{p \in C} n_p \cdot p$ is a divisor, $\deg D = \sum_{p \in C} n_p$.

Theorem 2.6: Riemann-Hurwitz Formula

Let C, C' be compact Riemann surfaces where:

$$\text{genus}(C) = g \quad \text{and} \quad \text{genus}(C') = g'$$

Let $f : C \rightarrow C'$ be a nonconstant holomorphic mapping of degree n . Let R be the ramification divisor of f . Then:

$$\deg R = 2(g + n - ng' - 1)$$

Proof :

First, we shall use the fact that on any compact Riemann surface, there exists a nontrivial meromorphic differential. This could be intuitively seen by the correspondence between compact Riemann surfaces and curves, and on any curve we have a non-trivial meromorphic differential.

Hence, choose $\omega \in K^1(C')$, and let $f^*\omega$ be its pullback. Locally we have:

$$\omega = z^\nu$$

and

$$\omega = g(\omega)d\omega$$

Hence:

$$f^*\omega = \nu g(z^\nu)z^{\nu-1}dz$$

and so the order of each point is:

$$\nu_{f^*\omega}(p) = \nu_f(p)\nu_\omega(f(p)) + \nu_f(p) - 1$$

and so the divisor of $(f^*\omega)$ is :

$$\begin{aligned} (f^*\omega) &= \sum_p \nu_f(p)\nu_\omega(f(p))p + R \\ &= \sum_q \nu_\omega(q) \left(\sum_{f(p)=q} \nu_f(p)p \right) + R \\ &= \sum_q \nu_\omega(q)f^{-1}q + R \\ &= f^{-1} \left(\sum_q \nu_\omega(q)q \right) + R = f^{-1}((\omega)) + R \end{aligned}$$

Thus:

$$\deg(f^*\omega) = \deg f^{-1}((\omega)) + \deg R$$

Then from the Poincaré-Hopf formula:

$$\deg(f^*\omega) = 2g - 2 \quad \deg(\omega) = 2g' - 2$$

Thus directly computing we get:

$$\begin{aligned} \deg f^{-1}(\omega) &= \sum_q \nu_\omega(q) \left(\sum_{f(p)=q} \nu_f(p) \right) \\ &= \deg(\omega) \deg f \\ &= (2g' - 2)n \end{aligned}$$

then substituting we get:

$$2g - 2 = (2g' - 2)n + \deg R$$

re-arranging gives our desired result.

From this we can immediately conclude:

Corollary 2.7: Holomorphic Map and Genus

Let $f : C \rightarrow C'$ be a holomorphic map between genus surfaces of genus g, g' . Then if $n \geq 1$:

$$g \geq g'$$

Proof :

Apply the formula.

Corollary 2.8: Compact Riemann Surface As Algebraic Equation

Let C be a compact Riemann surface and let

$$x, y : C \rightarrow \mathbb{CP}^1$$

be holomorphic mapping where $\deg x = d$. Then there exists a polynomial equation f such that $f(x, y) = 0$.

Proof :

[2, p.94]

Next, we cover the genus formula.

Theorem 2.9: Genus Formula

Let $C \subseteq \mathbb{CP}^2$ be an irreducible algebraic curve of degree d with δ singular points all being ordinary double points, and let $\sigma : \tilde{C} \rightarrow C$ be its normalization where $\text{genus}(\tilde{C}) = g$. Then:

$$g = \frac{(d-1)(d-2)}{2} - \delta$$

Proof :
[2, p.95].

3 The Riemann-Roch Theorem

HERE

References

- [1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.
- [2] Phillip A. Griffiths. *Introduction to Algebraic Curves*. Vol. 78. Mathematical Monographs. American Mathematical Society.
- [3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.